Basic Principles of Medicine 1

Module: Foundation to Medicine| DLA 20 before FTM LEC 37

Introduction to the Nervous System Part 2 - CNS and PNS

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Objectives

SOM.MKII.BPM1.1.FTM.1.ANAT.1011	Describe the basic organization and function of the brain and cranial nerves
SOM.MKII.BPM1.1.FTM.1.ANAT.1012	Describe the basic organization and function of the spinal cord and spinal nerves

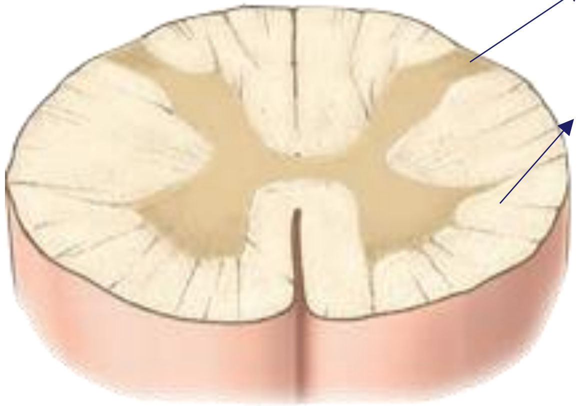
Components of the Nervous System

- Central Nervous System
  - White matter is composed of myelinated tracts of axons
  - Gray matter is composed of neuronal cell bodies
  - Collection of cell bodies is called a nucleus
  - Collection of axons is called a tract

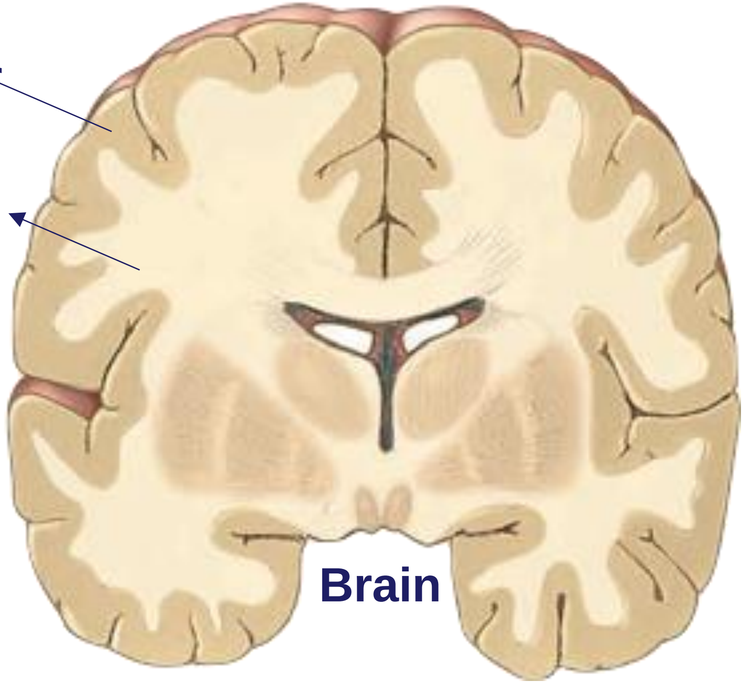
- Peripheral Nervous System
  - Collection of cell bodies is called a ganglion
  - Collection of axons is called a peripheral nerve

grey matter

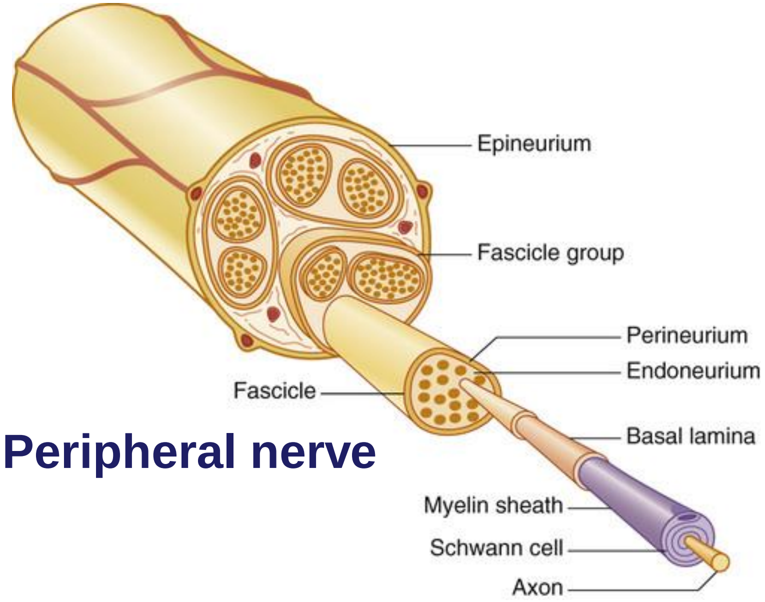
white matter



Spinal cord



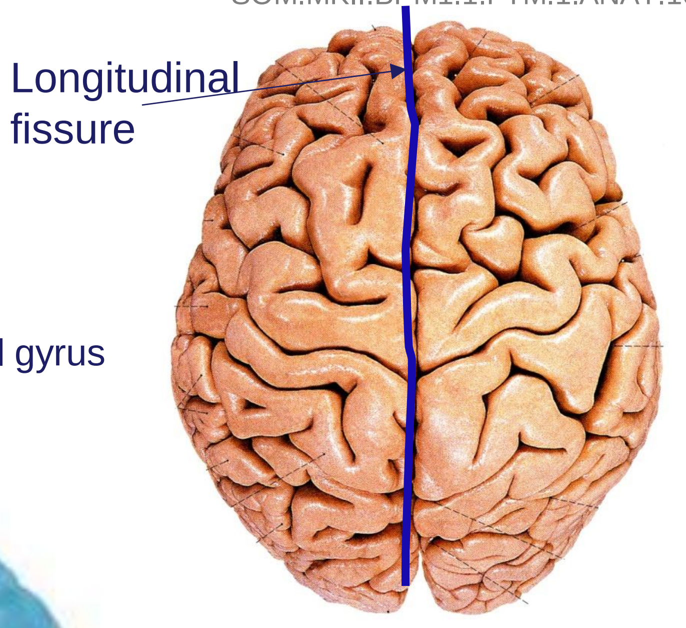
Brain

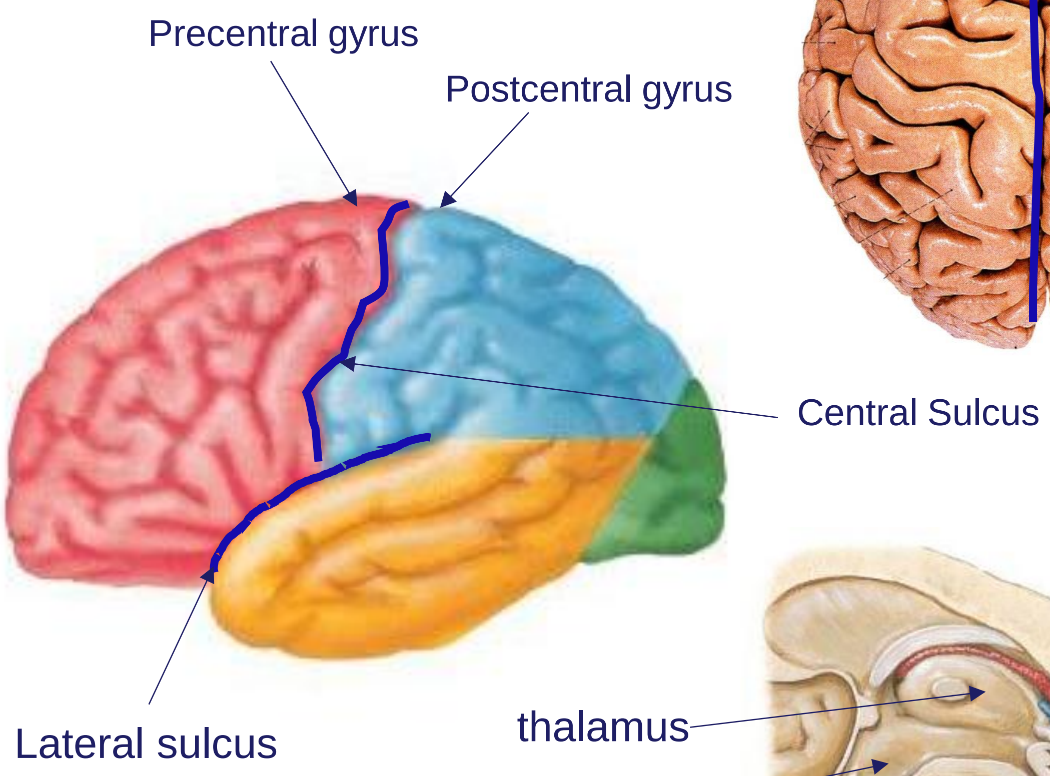


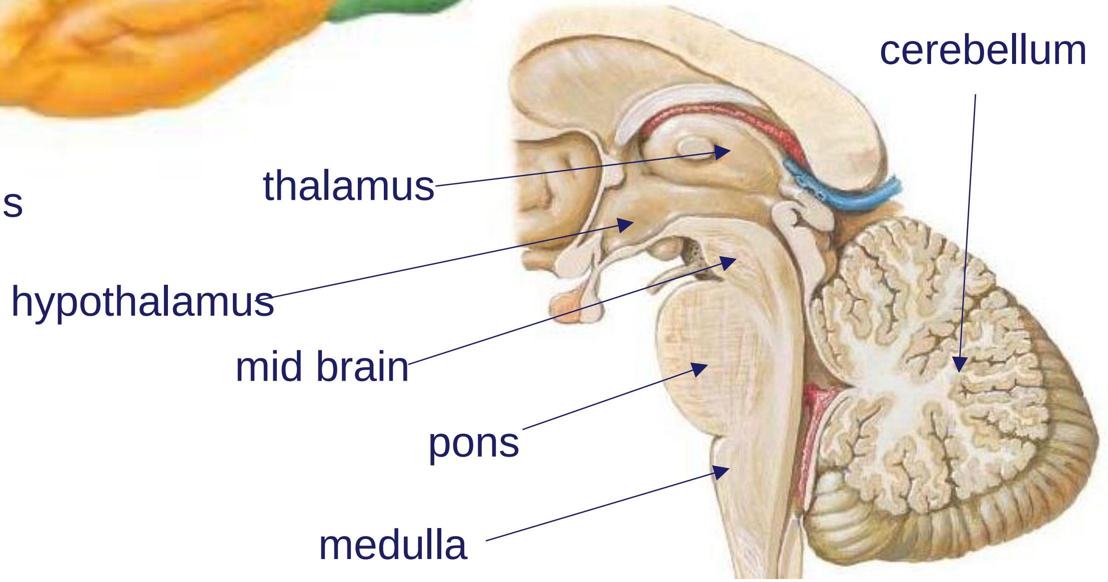
Peripheral nerve

CNS – Brain

- Cerebral Hemispheres
  - Surface area of the brain is larger than the space in the cranial cavity which causes folding
  - Folds = gyri are separated by grooves = sulci
  - Deep groove called longitudinal fissure – separates left and right cerebral hemispheres
  - Specific sulci separates lobes
  - Generalized functions of lobes
    - sensory, motor, visual, auditory, language and higher cognitive functions, memory etc.
- Thalamus
  - relaying and modulating sensory input to cerebrum
  - modulating cerebral activity to regulate motor function
- Hypothalamus
  - homeostasis, growth and reproduction
- Brainstem:
  - Midbrain, Pons and Medulla Oblongata
  - Vegetative functions such as cardio regulation, swallowing, yawning etc.
- Cerebellum
  - posture, balance and smooth coordinated movements





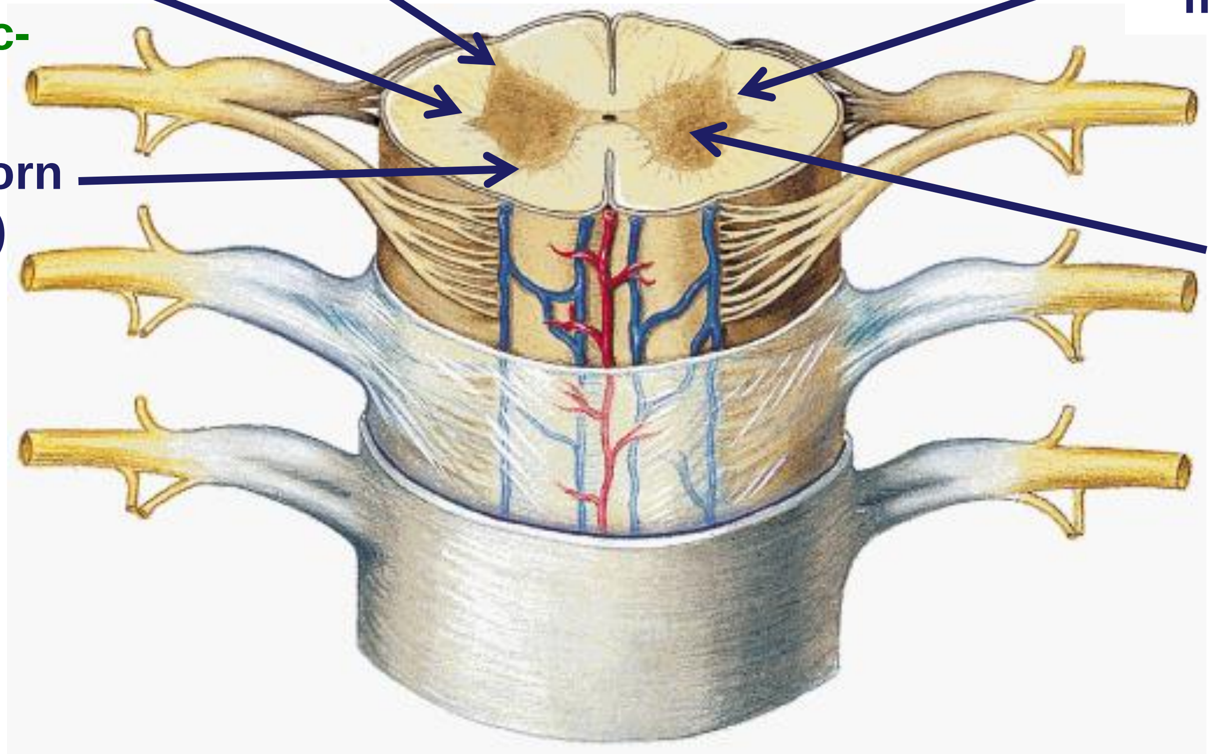


PNS – Cranial Nerves

- Axons of neurons forming peripheral nerves exiting the cerebral cortex and brainstem
- **12 pairs**
  - **Cortex- CN I - CN II**
  - **Midbrain – CN III - CNIV**
  - **Pons – CNV - CNVIII**
  - **Medulla- CN IX – XII**
- Can consist of motor, sensory, parasympathetic fibers

CNS - Spinal Cord: Internal Organization

- Made up of two types of tissue:
  - Inner gray matter (nuclei)– cell bodies of neurons and glial cells
  - outer white matter (tracts)– myelinated axons, neural "highways"
- Segments
  - cervical, thoracic, lumbar, sacral and coccygeal

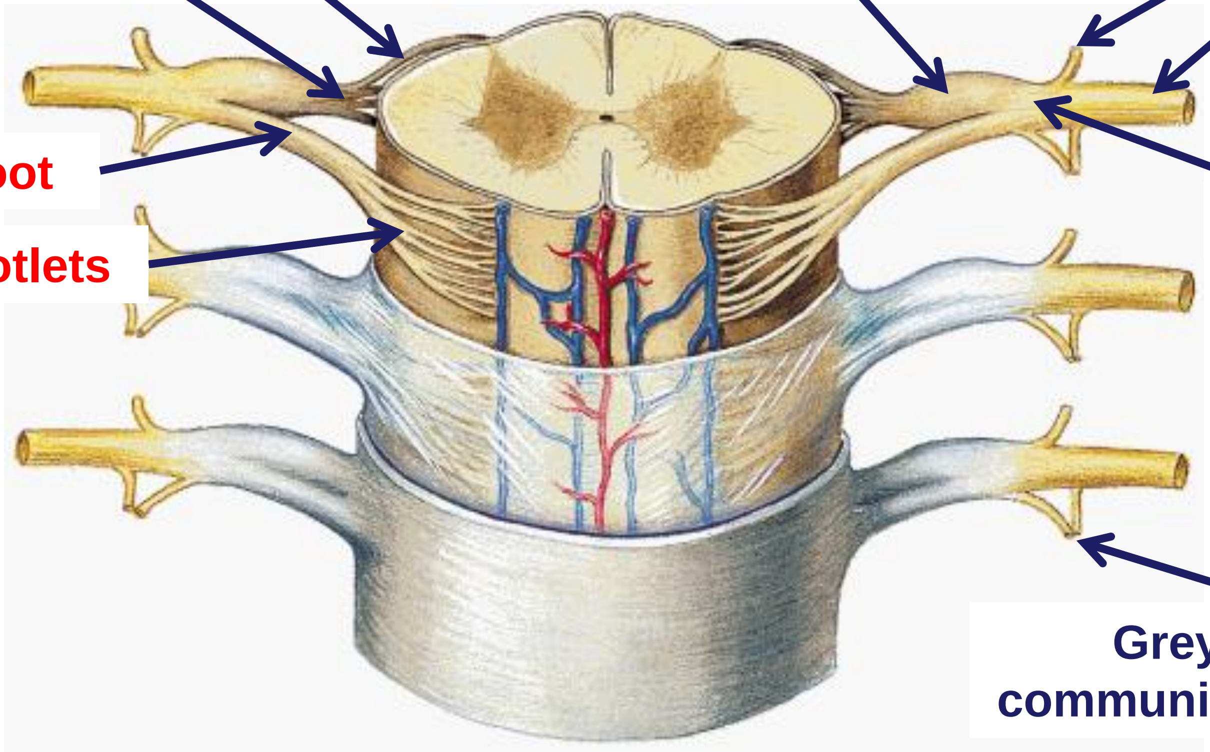


Formation of a typical spinal nerve

Dorsal horn (sensory) — Dorsal rootlets (sensory) — Dorsal root (sensory)

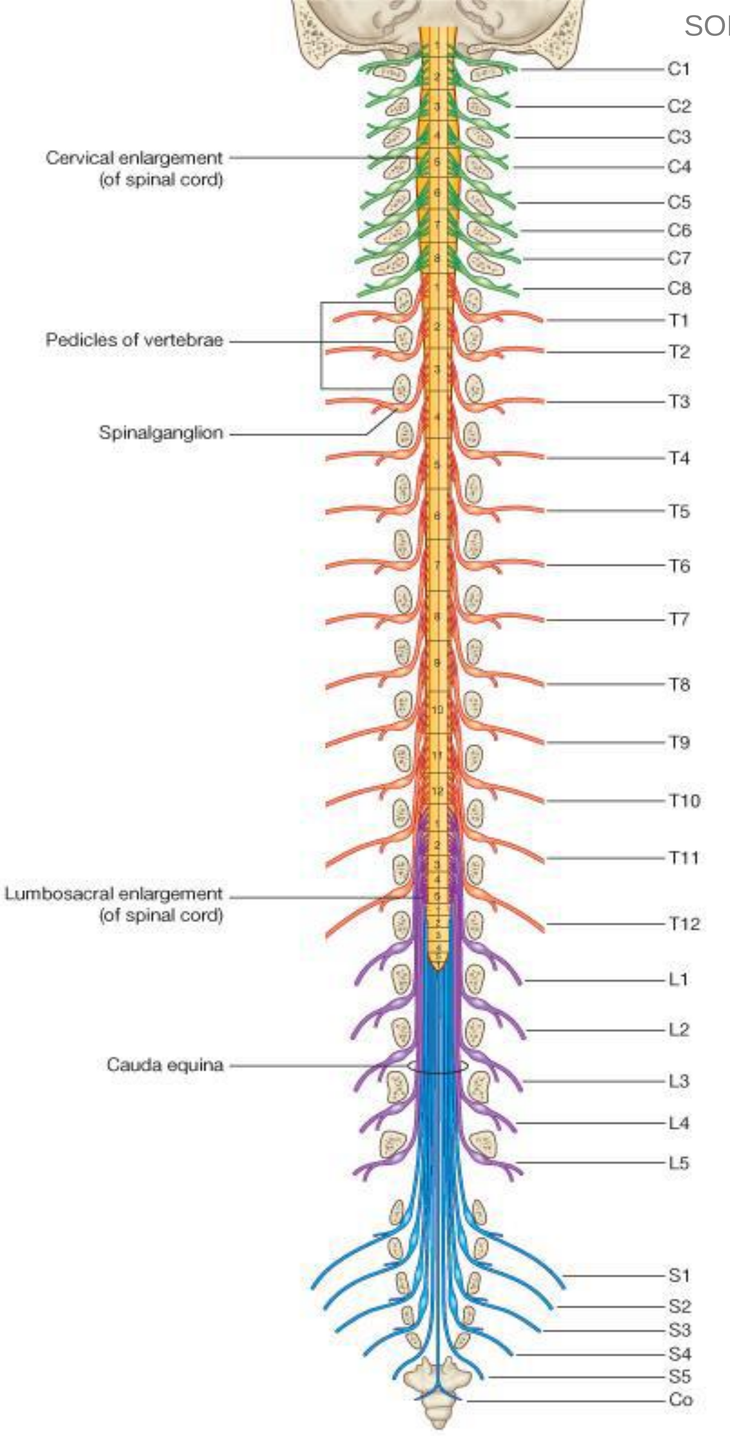
Ventral horn (motor) — Ventral rootlets (motor) — Ventral root (motor)

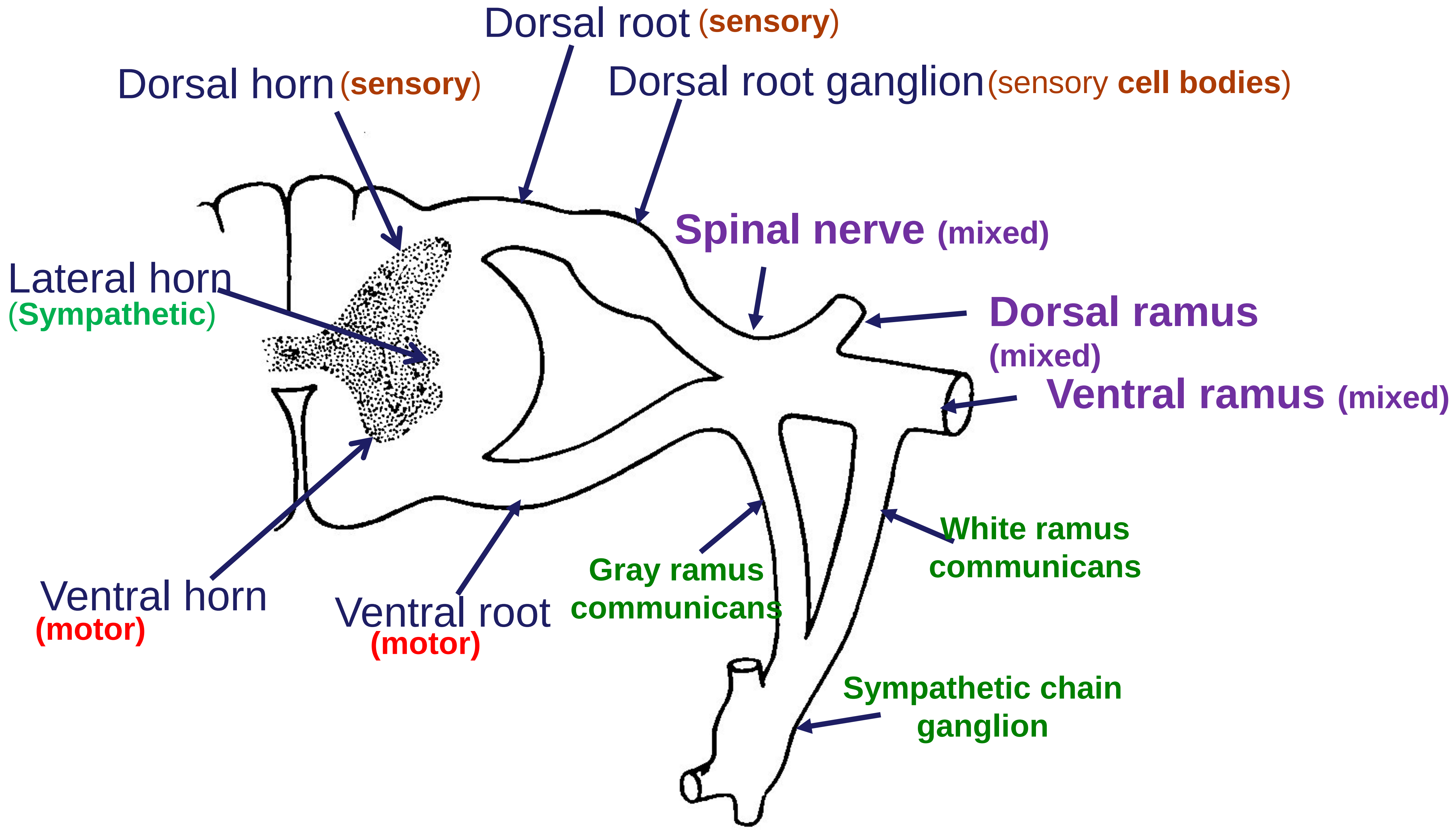
Dorsal root + Ventral root → Spinal nerve (mixed) → Dorsal ramus (mixed), Ventral ramus (mixed)



PNS - Spinal Nerves

- Axons of neurons forming peripheral nerves exiting the spinal cord segments
- 31 pairs
  - **8 cervical (C1 – C8)**
  - **12 thoracic (T1 – T12)**
  - **5 lumbar (L1 – L5)**
  - **5 sacral (S1 – S5)**
  - **1 coccygeal (Co1)**
- Can consist of motor, sensory, sympathetic and parasympathetic fibers




Dorsal root (sensory)
Dorsal horn (sensory)
Dorsal root ganglion (sensory cell bodies)
Spinal nerve (mixed)
Lateral horn
(Sympathetic)
Dorsal ramus
(mixed)
Ventral ramus (mixed)
White ramus
communicans
Gray ramus
communicans
Ventral horn
(motor)
Ventral root
(motor)
Sympathetic chain
ganglion

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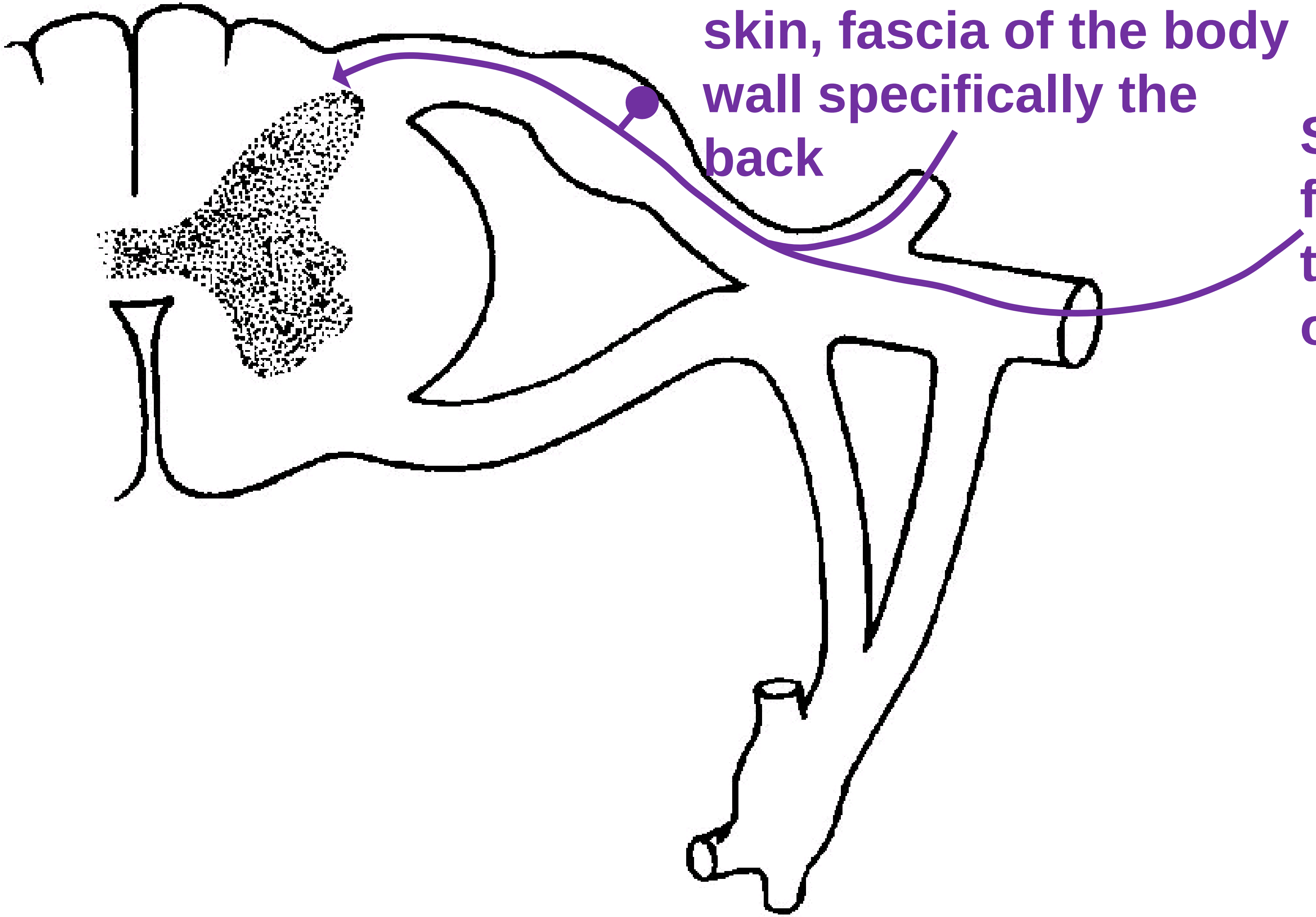
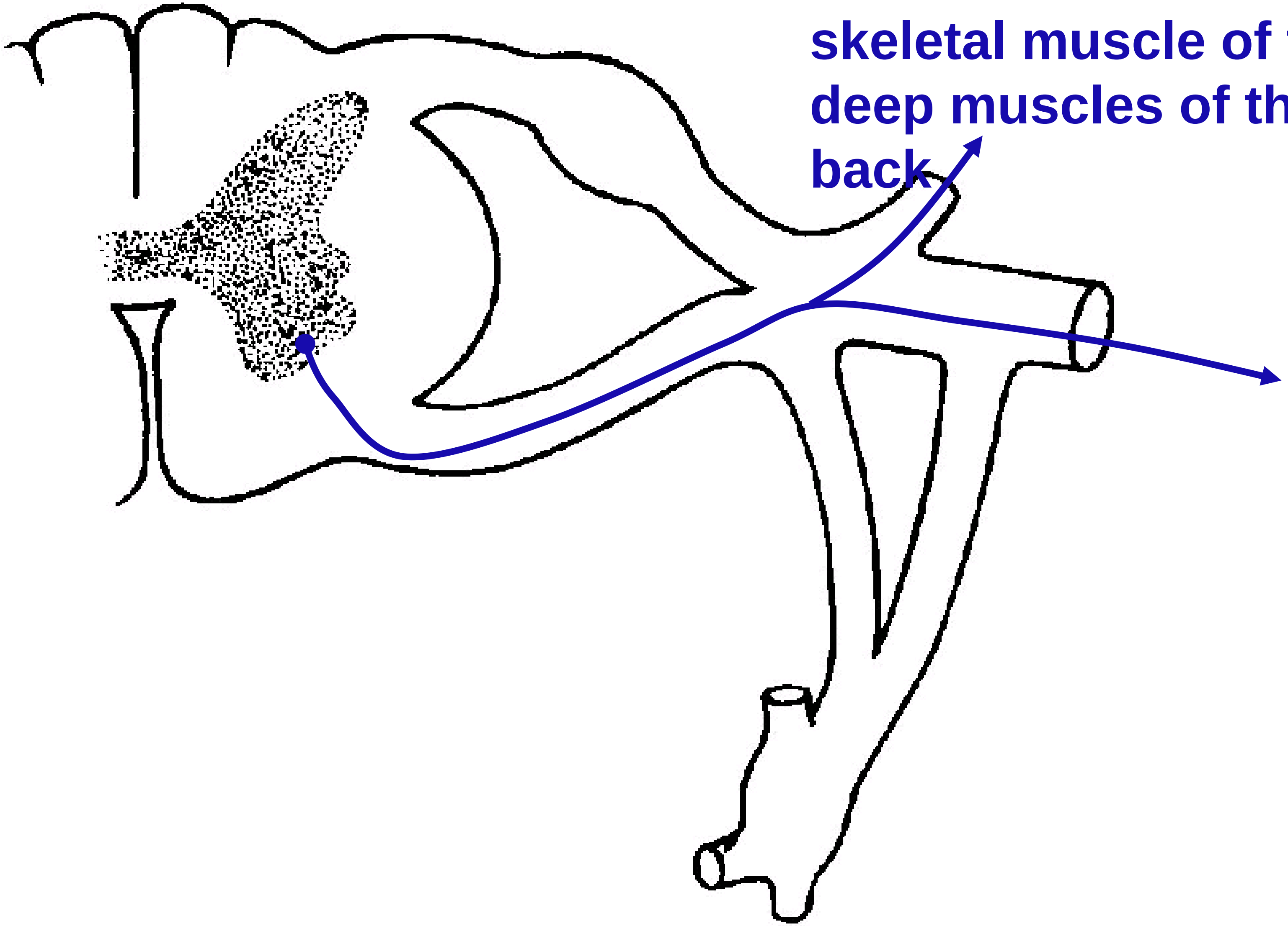
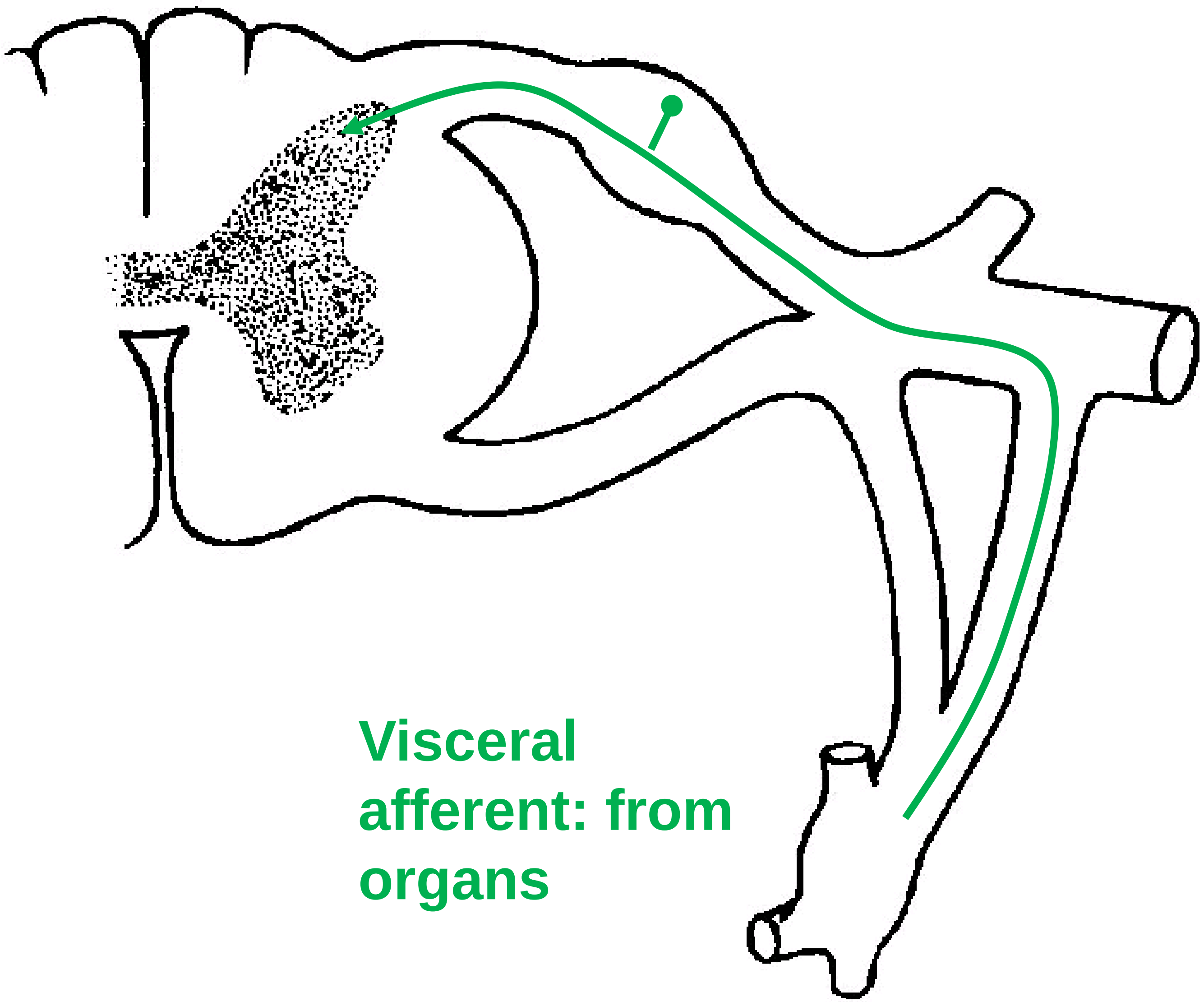
Spinal Cord and Spinal Nerve Organization



Spinal Cord and Spinal Nerve Organization

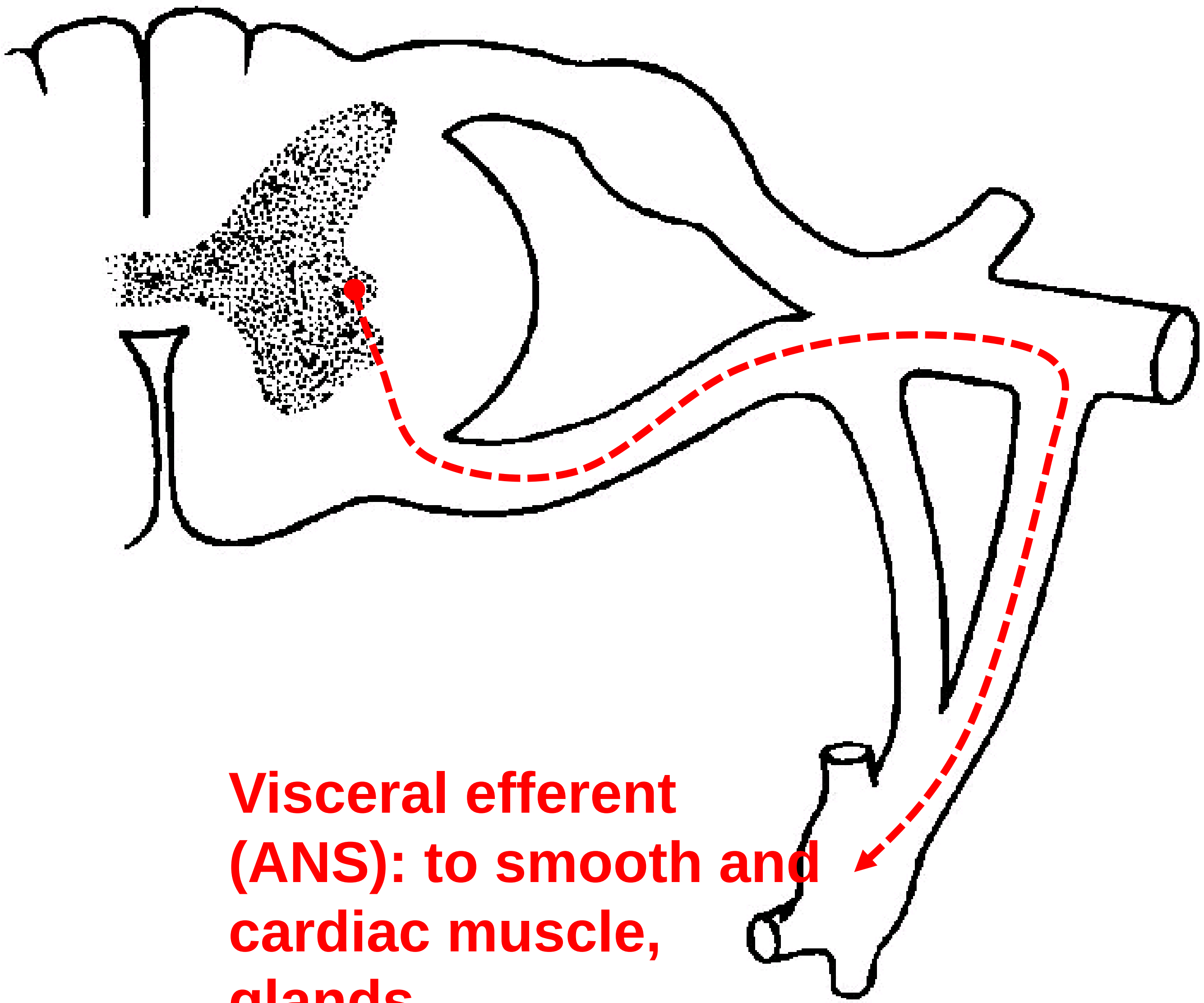


Spinal Cord and Spinal Nerve Organization



Visceral afferent: from organs

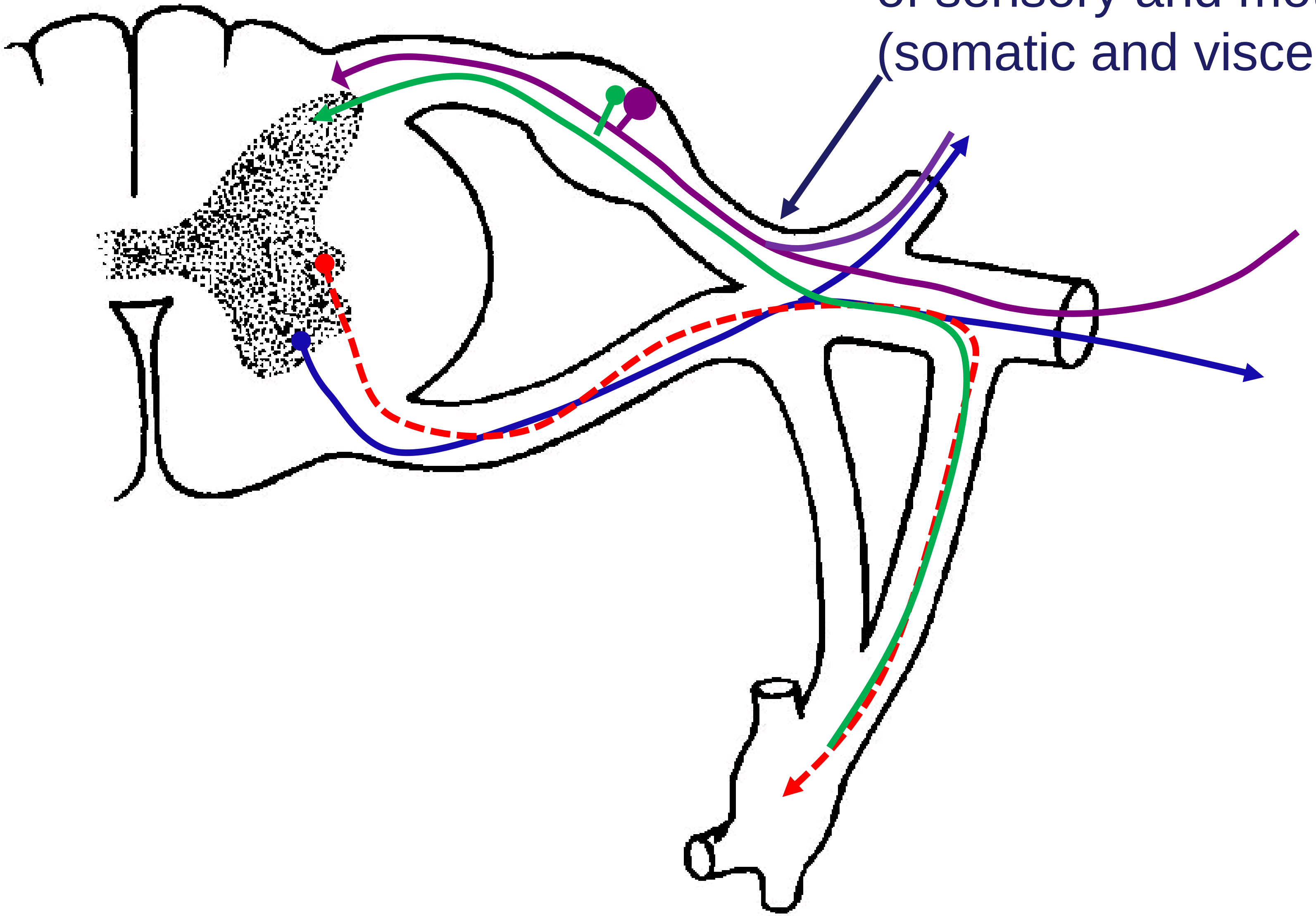
Spinal Cord and Spinal Nerve Organization



Visceral efferent (ANS): to smooth and cardiac muscle, glands

Spinal Cord and Spinal Nerve Organization

A spinal nerve contains a mixture of sensory and motor fibers (somatic and visceral)



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